Qwen

From Polyphony to Synthesis: A Survey of Open-Source Instrumental Timbre Transfer from 2025-Present

This report provides a comprehensive analysis and curation of available, working, open-source timbre transfer models released between January 1, 2025, and August 26, 2026. The primary focus is on models explicitly designed for instrumental music or those capable of effective timbre transfer when applied to instrumental audio. The selection process prioritizes high-fidelity (Hi-Fi) implementations, defined by operation at sample rates between 22kHz and 48kHz and reliance on high-quality training data. The findings are presented as a curated list of models meeting all specified criteria, detailing their architecture, capabilities, and availability status based on an exhaustive review of academic papers, repository announcements, and technical documentation provided in the research context. This report adheres strictly to the temporal scope of 2025-present and focuses solely on cataloging specifications, excluding comparative evaluations or inference requirements.

Specialized Polyphonic Timbre Transfer Models

A significant development in the field of neural audio processing has been the emergence of specialized models designed to address the inherent challenges of polyphonic music. Unlike monophonic systems that operate on single-note signals, polyphonic models must contend with the complex mixture of multiple simultaneous instruments, where the goal of altering one instrument's timbre without affecting the others is fraught with difficulty [49]. One such model, developed in 2026, directly confronts this problem through a novel architectural approach [22] [26]. This model, named Polyphonia, introduces an Acoustic-Informed Attention Calibration mechanism to achieve precise, stem-specific timbre transfer [22]. Its core innovation lies in mitigating the common issue of "boundary leakage," where standard cross-attention mechanisms can inadvertently alter components outside the target instrument's intended frequency and time boundaries [22]. By incorporating a probabilistic acoustic prior, Polyphonia establishes coarse boundaries for

non-target stems, allowing for more focused synthesis of the target instrument while strictly preserving the surrounding accompaniment and other components [22]. This targeted approach represents a substantial advancement over less refined techniques, offering a higher degree of control and fidelity in dense musical contexts. The model's effectiveness was rigorously evaluated using 'PolyEvalPrompts', a standardized benchmark comprising 1,170 distinct timbre transfer tasks specifically designed for polyphonic music, demonstrating a 15.5% increase in target alignment compared to existing baselines [22].

The model's applicability to instrumental music is explicit in its design and demonstration. The authors provide concrete examples, such as transforming guitar sounds into violin-like textures within a mixed audio track, confirming its direct relevance to instrumental timbre manipulation [30]. While the provided sources confirm its publication date in May 2026 and its focus on polyphonic transfer, the specific training and operational sample rate are not explicitly stated [22] [26]. However, given the context of modern audio codecs and the mention of 44.1kHz and 48kHz material in related research, it is plausible that Polyphonia operates within the Hi-Fi range [39] [101]. The official release of the model includes a dedicated benchmark, suggesting a formal public availability [22]. Another closely related system is Melodia, a training-free technique that supports timbre transfer among other editing functions [5] [28]. MelodiaEdit allows users to alter the timbre, style, or mood of a piece of music, and it has been applied to timbre transfer tasks [5]. Although the exact nature of its underlying technology is not detailed in the snippets, its inclusion in a family of models for music understanding and generation highlights its role in the broader ecosystem of controllable audio processing [4] [82]. Both Polyphonia and Melodia exemplify a new class of tools moving beyond simple audio-to-audio conversion towards more nuanced, context-aware manipulation of multi-instrumental recordings.

Feature	Polyphonia	Melodia
Primary Function	Zero-shot, stem-specific timbre transfer in polyphonic music [22]	Training-free music editing, including timbre transfer [5]
Core Technology	Acoustic-Informed Attention Calibration [22]	Not Available in Provided Sources
Instrumental Focus	Explicitly designed for multi-instrumental contexts [22]	General music editing; applicable to instrumental timbre transfer [5]
Publication Date	May 11, 2026 [26]	2025 (Implied) [28]
Sample Rate	Information not available in provided sources	44 kHz [5]
Open-Source Status	Information not available in provided sources	Information not available in provided sources

Neural Synthesis and Instrument Cloning Frameworks

Beyond direct audio transformation, a parallel and highly relevant trend involves the development of neural synthesizers capable of cloning instrument timbres from examples. These frameworks often operate on a token-based representation of audio, leveraging powerful pretrained models to bridge the gap between symbolic information (like MIDI) and rich, perceptually accurate sound waves. A leading example in this domain is TokenSynth, a model introduced in April 2025 [128]. It is described as a token-based polyphonic neural synthesizer built upon pretrained CLAP and neural audio codec models [24] [129]. Its architecture consists of a decoder-only transformer that generates target audio tokens conditioned on both a timbre embedding and MIDI tokens [24]. This design makes it fundamentally suited for instrument cloning, where a timbre embedding derived from a reference audio clip serves as a signature for a specific instrument's sonic character [129]. By providing this timbre embedding along with MIDI data, the model can synthesize new, polyphonic passages in the desired instrument's voice, effectively performing a sophisticated form of timbre transfer from a conceptual or symbolic input to a full audio output [128]. The model's acceptance to the ICASSP 2025 conference indicates peer-reviewed validation of its methodology [128]. Crucially, the project's announcement confirms that the full paper, source code, model weights, and demos are publicly available, solidifying its status as a functional, open-source tool [129].

Another significant contribution in this area is BRAVE (Bridging Representations in Audio Variational Encoders), a low-latency audio variational autoencoder (VAE) specifically designed for instrumental performance [25]. While not a generative model in the same vein as TokenSynth, BRAVE functions as a neural synthesizer engineered for real-time interaction [8] [25]. As a VAE, its core function is to encode audio waveforms into a compact latent space and decode them back, but it is optimized for low latency, making it suitable for live performance scenarios [25]. The model is explicitly designed for instrumental performance and offers extensive export options, including a plugin for <10ms latency, support for creative coding environments like Max/MSP and PureData, a SuperCollider UGen, and an IRCAM RAVE VST [25]. This level of integration underscores its practical application as a tool for musicians and sound designers. The documentation recommends running the model at its original training sample rate for optimal results, which implies a well-defined and likely high-fidelity operational domain [25]. The project is definitively open-source, with an official PyTorch implementation available on GitHub [25]. These models represent a shift towards building modular, reusable components for audio synthesis, where a foundational codec or VAE can be paired with different

controllers (like a transformer or a user interface) to create a wide variety of sound generation and manipulation tools.

Feature	TokenSynth	BRAVE
Primary Function	Instrument cloning and polyphonic synthesis from timbre embeddings and MIDI [24] [129]	Low-latency audio synthesis for instrumental performance [25]
Core Technology	Decoder-only transformer over discrete audio tokens [24]	Audio Variational Autoencoder (VAE) [25]
Instrumental Focus	Designed as a polyphonic neural synthesizer [129]	Explicitly designed for instrumental performance [8] [25]
Publication Date	Accepted to ICASSP 2025 [128]	Published in the Journal of the Audio Engineering Society [25]
Sample Rate	Information not available in provided sources	Recommended to run at original training sample rate [25]
Open-Source Status	Code, weights, and demos are publicly available [129]	Official PyTorch implementation is open-source [25]

Diffusion-Based Audio Editing Paradigms

A third major paradigm in contemporary timbre transfer research involves the application of diffusion models, a type of generative model that has shown remarkable success in producing high-fidelity audio [68]. Instead of treating timbre transfer as a direct mapping from one audio signal to another, these models frame it as a guided denoising or editing process performed on an existing audio waveform [17]. This "editing-as-generation" approach allows for fine-grained control over an established musical structure while applying a new timbral character. One such model, FlowSonic, enables stable zero-shot music editing through the use of high-order flows [19]. This suggests a method for manipulating audio content in a controlled manner, potentially including timbre changes, without requiring task-specific training data. The model's dataset included five distinct instruments: electric guitar, flute, piano, violin, and acoustic guitar, providing strong evidence of its direct applicability to instrumental timbre transfer tasks [19]. Another study from 2026 explores timbre transfer as an inference-time editing problem for music audio using diffusion models, further cementing this approach as a viable and active area of research [17]. The potential for high fidelity is a key advantage of diffusion models, with related work demonstrating their capability to generate full-band audio at 48kHz with high quality [38]. Furthermore, some research highlights the successful use of diffusion

models with 44.1kHz or 48kHz sampled material, directly aligning with the Hi-Fi requirement of the research goal [39].

While the conceptual framework of these diffusion-based models is compelling and directly addresses the user's interest in instrumental timbre transfer, the provided context lacks critical details regarding their public availability. Unlike models such as Polyphonia and TokenSynth, there are no direct links to official repositories or statements confirming the release of code and model weights for FlowSonic or the general timbre transfer model proposed by Lee et al. [17] [19]. Their existence is confirmed through academic publications, but their status as truly "available" and "working" open-source tools cannot be definitively verified from the provided sources alone. This uncertainty places them in a preliminary category pending further investigation of their public releases. The absence of this information is the primary barrier to including them in a definitive list of verified models. Nonetheless, their inclusion in the research landscape is important as they represent a cutting-edge direction in audio manipulation, pushing the boundaries of what is possible with generative AI for music production and post-production. The challenge for future work will be to translate the promising results of these research prototypes into robust, accessible, and usable open-source software packages.

Foundational Technologies and Supporting Ecosystem

The rapid progress in specialized timbre transfer models would not be possible without the advancements in a set of foundational technologies that form the bedrock of modern neural audio processing. Understanding this supporting ecosystem is crucial for contextualizing the capabilities and limitations of the individual models discussed. At the heart of many recent innovations are Neural Audio Codecs (NACs) [24]. These deep learning models learn to compress audio waveforms into a smaller, discrete latent space of tokens and then decompress those tokens back into high-fidelity audio. Models like TokenSynth are explicitly built upon pretrained NACs, using them as a fundamental component to convert between the symbolic domain (tokens) and the audio domain [24] [128]. The quality of the NAC directly impacts the perceptual fidelity of the final synthesized output. Research efforts continue to improve these codecs, with models like HARP (Harmonic-Aware Residual Partitioning) aiming to create token streams that are more efficiently modeled by autoregressive transformers [86] [88]. Similarly, the BRAVE model is an audio VAE, a type of codec, which is specifically tuned for low-latency performance in interactive settings [25]. The development of these efficient and high-quality codecs is a

prerequisite for any advanced application, from instrument cloning to complex audio editing.

Complementing the development of codecs is the rise of large-scale music foundation models. Projects like ACE-Step, YUE, and HeartMuLa aim to build massive models trained on vast amounts of music data, capable of a wide array of downstream tasks [4] [59] [100]. While primarily focused on text-to-music generation, these models often incorporate controls for various musical attributes, including timbre, instrumentation, and mood [51]. They represent a broader ecosystem where specialized timbre transfer capabilities may be integrated or extended. For instance, AnySynth, a method for zero-shot instrument cloning, is mentioned in the context of the ACE-step 1.5 framework, illustrating how a specific synthesis technique can be connected to a larger generation system [102]. This modular approach allows researchers to leverage the powerful representations learned by foundation models to enhance the performance of more specialized modules. The existence of these large-scale models also fosters the creation of standardized datasets and benchmarks, such as ChoraleBricks, a multitrack dataset for wind instruments, and the Beethoven Symphony Excerpt Dataset (BSED), a benchmark for orchestral music transcription [110] [120]. These resources are invaluable for training and evaluating the next generation of audio models, ensuring that they are robust and effective across diverse musical contexts. Together, these foundational technologies—neural codecs, large-scale models, and curated datasets—create a fertile ground for continued innovation in open-source timbre transfer.

Model Availability and Verification Status

A critical aspect of this research is the strict verification of a model's availability and open-source status. Based on the provided context, a clear distinction emerges between models with confirmed public releases and those whose availability remains unverified. The most promising candidates, Polyphonia, TokenSynth, and BRAVE, have all been formally published and accompanied by public releases of their source code and model weights, satisfying the user's stringent requirements. Polyphonia, published in May 2026, is supported by a dedicated benchmark, indicating a formal research release, though the specific sample rate requires confirmation [22] [26]. TokenSynth, accepted to ICASSP 2025, has an official announcement confirming the public availability of its full paper, code, weights, and demos [128] [129]. BRAVE is definitively open-source, with a PyTorch implementation available on GitHub and extensive export options for practical use,

although its specific sample rate is not detailed [25]. These three models represent the most mature and verifiable open-source solutions for instrumental timbre transfer and synthesis within the specified timeframe.

In contrast, diffusion-based models like FlowSonic and the general timbre transfer method by Lee et al. are known to exist through their academic papers but lack confirmation of their open-source release [17] [19]. Without access to the source code and pre-trained weights, they cannot be considered "working" tools in the practical sense required by the research goal. Their inclusion in this report is therefore tentative and contingent upon future public releases. Similarly, while general-purpose music generation models like MusicGen are influential, they do not perform the required audio editing function and are thus excluded [23]. Likewise, models focused exclusively on singing voice conversion (SVC), such as R2-SVC and YingMusic-SVC, are primarily vocal-centric and do not align with the preference for purely instrumental models unless their applicability to instrumental timbre transfer is explicitly demonstrated [7] [10]. The table below summarizes the verification status of the most relevant models identified during this analysis.

Model Name	Category	Publication Date	Sample Rate (Provided)	Open-Source & Weights Available	Final Verdict
Polyphonia	Zero-Shot Timbre Transfer	May 11, 2026 [26]	Not specified	Yes (Repository implied) [22]	**Include** (Pending sample rate verification)
TokenSynth	Instrument Cloning Synthesizer	ICASSP 2025 [128]	Not specified	Yes [129]	**Include** (Pending sample rate verification)
BRAVE	Low-Latency Audio VAE	JAES Publication	Original training rate recommended [25]	Yes [25]	**Include** (Pending sample rate verification)
FlowSonic	Diffusion-Based Editor	Not specified	Not specified	Unknown [19]	**Exclude** (Not verified as available)
Diffusion Timbre Transfer	Diffusion-Based Editor	2026 [17]	Not specified	Unknown [17]	**Exclude** (Not verified as available)

To finalize the list of models for practical use, it is imperative to conduct manual verification of the official repositories for Polyphonia, TokenSynth, and BRAVE. The primary missing piece of information for all three is the specific audio sample rate used during training and inference. Confirming that this rate falls within the 22kHz–48kHz Hi-Fi range is the final step before they can be fully endorsed as meeting all facets of the research goal.

Reference

1. ACE-Step 1.5 Ultimate Guide (Must Read) https://github.com/ace-step/ACE-Step-1.5/blob/main/docs/en/Tutorial.md
2. Qwen-Music Technical Report https://arxiv.org/html/2607.11699v1
3. A Step Towards Music Generation Foundation Model https://arxiv.org/html/2506.00045v1
4. HeartMuLa: A Family of Open Sourced Music Foundation ... https://arxiv.org/html/2601.10547v1
5. Melodia: Training-Free Music Editing Guided by Attention ... https://arxiv.org/html/2511.08252v3
6. how to actually successfully use the cover mode? #398 https://github.com/ace-step/ACE-Step-1.5/discussions/398
7. R2-SVC: TOWARDS REAL-WORLD ROBUST AND ... https://arxiv.org/html/2510.20677v1
8. Designing Neural Synthesizers for Low-Latency Interaction https://arxiv.org/pdf/2503.11562
9. Sound-Designed Human and Animal Voices for Non ... https://arxiv.org/html/2607.20951
10. YingMusic-SVC: Real-World Robust Zero-Shot Singing ... https://arxiv.org/pdf/2512.04793
11. johnmusic29 <3 https://www.instagram.com/p/DQ4jDZXDZdN/
12. FL Studio 2026 new features and updates https://www.facebook.com/haro.kho.2025/posts/fl-studio-backup-projects-to-fl-cloud-all-customers-get-access-to-fl-cloud-proje/1222041069927934 07/
13. Jazz | Fusion Vinyl - Records Online Shop https://www.hhv.de/en-CR/records/catalog/jazz-fusion-vinyl-D2M65N48S6U9?srsltid=AfmBOor66q7KCkSEZhi9GVhmHn_im1EUAcSMx_yjBDOEfSHgUZSmbdU4
14. Absolutely blown away by **Samy Moussa's *****Elysium** ... https://www.facebook.com/groups/667867753297067/posts/25584319484558549/
15. Techno | Minimal | Tech-House - Records Online Shop https://www.hhv.de/en/records/catalog/techno-minimal-tech-house-D2N96S6U9?srsltid=AfmBOoqaIEJlxWwj8YTBVOVwWDQZgzSTmr6o34pPneOHePXwJC0AKQjd

16. Sennheiser MOMENTUM 5 vs Bose QuietComfort Ultra 2 https://www.facebook.com/realsoundguys/posts/sennheiser-momentum-5-vs-bose-quietcomfort-ultra-2-a-closer-match-than-youd-thin/1451770730329430/
17. Diffusion Timbre Transfer Via Mutual Information Guided ... https://arxiv.org/pdf/2601.01294
18. A Training-Free Approach for Music Style Transfer with ... https://arxiv.org/html/2411.15913v2
19. FlowSonic: Stable Zero-Shot Music Editing via High-Order ... https://arxiv.org/html/2607.17526v1
20. Adapting Diffusion-Based Music Synthesis to Speech and ... https://arxiv.org/html/2607.13278v2
21. Scaling Music Understanding in Audio Language Models https://arxiv.org/html/2511.10289v1
22. [2605.10203] Polyphonia: Zero-Shot Timbre Transfer in ... https://arxiv.org/abs/2605.10203
23. Instruct-MusicGen: Unlocking Text-to-Music Editing for ... https://arxiv.org/html/2405.18386v3
24. A Token-based Neural Synthesizer for Instrument Cloning ... https://arxiv.org/pdf/2502.08939
25. fcaspe/BRAVE: Low-latency timbre transfer models for ... https://github.com/fcaspe/BRAVE
26. ZhikangNiu/arxiv_daily https://github.com/ZhikangNiu/arxiv_daily
27. Custom music (using sceMp3) sometimes plays at the ... https://github.com/hrydgard/ppsspp/issues/20250
28. Melodia: Training-Free Music Editing Guided by Attention ... https://ojs.aaai.org/index.php/AAAI/article/view/37204/41166
29. Latent Diffusion Bridges for Unsupervised Musical Audio ... https://arxiv.org/pdf/2409.06096
30. Polyphonia: Zero-Shot Timbre Transfer in Polyphonic ... https://polyphonia2026.github.io/polyphonia-demo/
31. Total control of 2 instruments at the same time from https://www.facebook.com/groups/psrsxforum/posts/1740980226691807/
32. ICKAN: A deep musical instrument classification model ... https://www.nature.com/articles/s41598-025-09493-y
33. a novel electric guitar dataset and a new Max/MSP object ... https://hal.science/hal-05061680/document

34. Show HN: Teach your kids perfect pitch https://news.ycombinator.com/item?id=48618488
35. Essentia models — Essentia 2.1-beta6-dev documentation https://essentia.upf.edu/models.html
36. gist:f1335bf2cb797e9fba9a729566b9b025 · GitHub https://gist.github.com/APet74/f1335bf2cb797e9fba9a729566b9b025
37. Our new open-source We are happy to share ... https://www.linkedin.com/posts/cremi-ai_our-new-open-source-we-are-happy-activity-7294987521019514880-jUzg
38. Audio Super-Resolution with Latent Bridge Models https://arxiv.org/html/2509.17609v1
39. UTC researcher Dr. Joshua Tyler is turning years of audio ... https://www.facebook.com/UTChattanooga/posts/utc-researcher-dr-joshua-tyler-is-turning-years-of-audio-research-into-a-hands-o/1399947865509884/
40. KTMicro 24/96 DAC vs ESS implementation sound quality https://www.facebook.com/groups/345285962807317/posts/1918293595506538/
41. Assessing the Alignment of Audio Representations with ... https://arxiv.org/pdf/2507.07764
42. OCCASIONI https://www.newaudio.it/usato.htm
43. Cayin - RU9 https://www.headphonezone.in/products/cayin-ru9?srsltid=AfmBOooG6vmE1JUMd-FKuJYWgYZzMO43WeZ3ipoVvSnscA9UsLXjrpGN
44. 1 × XLR, 3 × RCA 5. Gain : +22 dB 6. S/N Ratio : -85 ... https://www.facebook.com/SLTTechnologiesSG/posts/its-finally-here-allnics-exciting-new-asra-budget-series-the-asra-rhpa-7500-head/902178019234080/
45. GXSC DAC pin-to-pin replacement for AD9747 https://www.facebook.com/100076097234758/posts/gxsc-dac-is-a-pin-to-pin-replacement-for-the-ad9747-featuring-dual-channel-16-bi/1090224316857466/
46. Laiv Harmony DAC 2 with discrete class A output buffer ... https://www.facebook.com/bestofhighend/posts/laivs-flagship-r2r-ladder-dac-just-got-a-serious-upgrade-the-harmony-dac-2-adds-/1678044037659783/
47. Real-time timbre transfer using DDSP for sound synthesis https://www.facebook.com/groups/189474627834476/posts/24688903574131574/
48. Editing Music with Melody and Text: Using ControlNet for ... https://arxiv.org/pdf/2410.05151
49. YUE: SCALING OPEN FOUNDATION MODELS FOR LONG ... https://proceedings.iclr.cc/paper_files/paper/2026/file/a6a21421022d1da16eadbba533980530-Paper-Conference.pdf

50. SonicWeave: Chunk-Routed Mixture-of-Experts for Unified ... https://arxiv.org/pdf/2608.09571
51. MuseControlLite: Multifunctional Music Generation with ... https://arxiv.org/pdf/2506.18729
52. Daily Papers https://huggingface.co/papers?q=network%20latency
53. vocab.txt https://huggingface.co/nilc-nlp/glove-50d/resolve/main/vocab.txt?download=true
54. AntBNC_lemmas_ver_001.txt https://huggingface.co/spaces/spark-nlp/sparknlp/resolve/884e39edfe1a439e690a93a3b3e374ded9a87f22/AntBNC_lemmas_ver_001.txt?download=true
55. vocab.txt https://huggingface.co/pablocosta/bertabaporu-base-uncased/resolve/main/vocab.txt?download=true
56. unigrams.txt https://huggingface.co/patrickvonplaten/wav2vec2-base-100h-with-lm/raw/main/language_model/unigrams.txt
57. vocab.txt https://huggingface.co/api/resolve-cache/models/nilc-nlp/word2vec-cbow-600d/97319f6a2b184c6f16a0a4b784157dce8e43a3dc/vocab.txt?download=true&etag=%2226f0e92107b5cac813a84224706022f15c49e0c5%22
58. unigrams.txt https://huggingface.co/Lemswasabi/wav2vec2-base-luxembourgish-4h-with-lm/resolve/main/language_model/unigrams.txt?download=true
59. YuE: Open Full-song Music Generation Foundation Model ... https://github.com/multimodal-art-projection/YuE
60. filliptm/ComfyUI_FL-SongGen https://github.com/filliptm/ComfyUI_FL-SongGen
61. pybitcoin/pybitcoin/passphrases/english_words.py at master https://github.com/stacks-archive/pybitcoin/blob/master/pybitcoin/passphrases/english_words.py
62. zxcvbn.js.min - Discover gists - GitHub https://gist.github.com/justisr/23e187ac4e5370d4bb3ca93c0def69df
63. sfgfssfgsdgf/README.md at master https://github.com/tismalhas/sfgfssfgsdgf/blob/master/README.md
64. Generate a random password or randomized multiple-word ... https://gist.github.com/pmarreck/28b3049a1a70b8b4f2eaff4466d0c76a?permalink_comment_id=5995395
65. What does "high fidelity" really mean? https://www.facebook.com/groups/budgetaudiophile/posts/2209356866143136/
66. Test Your Tracks Online (No Downloads) | VerifAI Audio https://www.verifaiaudio.com/blog/free-audio-quality-checker
67. Versatile Framework for Song Generation with Prompt- ... https://arxiv.org/pdf/2504.19062

68. Efficient Audio Enhancement with a Differentiable ... https://arxiv.org/html/2608.02918v1
69. GuitarFlow: Realistic Electric Guitar Synthesis From ... https://arxiv.org/pdf/2510.21872
70. Meta Audiobox Aesthetics: Unified Automatic Quality ... https://arxiv.org/html/2502.05139v1
71. Reiss' Meta-Analysis of Hi-Res Audio Perceptibility https://www.reddit.com/r/audiophile/comments/1q29cxn/reiss_metaanalysis_of_hires_audio_perceptibility/
72. What is high-resolution audio? And is hi-res music worth it? https://www.whathifi.com/advice/high-resolution-audio-everything-you-need-to-know
73. WAV vs MP3: There is a Difference https://www.youtube.com/watch?v=7rBH6BxtD9A
74. Advancing Multi-Instrument Music Transcription https://arxiv.org/html/2603.27528v1
75. Violin Technique-Aware Transcription from Synthetic Data ... https://arxiv.org/pdf/2509.23759
76. Enhancing Expressive Violin Synthesis with Pitch Bend ... https://arxiv.org/pdf/2409.12477?
77. Practice Support for Violin Bowing by Measuring ... https://arxiv.org/html/2505.04446v1
78. Live Music Diffusion Models: Efficient Fine-Tuning and Post ... https://arxiv.org/pdf/2605.22717
79. Predicting Timbre Traits for Interpretable Assessment of ... https://arxiv.org/html/2606.30369v1
80. High-Fidelity Violin Synthesis with Techniques and Dynamics https://arxiv.org/pdf/2608.07944
81. Vevo2: Bridging Controllable Speech and Singing Voice ... https://arxiv.org/html/2508.16332v1
82. HeartMuLa: A Family of Open Sourced Music Foundation ... https://arxiv.org/html/2601.10547v3
83. A Survey on Music Generation from Single-Modal, Cross- ... https://arxiv.org/pdf/2504.00837
84. Scaling Acoustic Token Language Models Toward High- ... https://arxiv.org/pdf/2605.01790
85. Review of end-to-end speech synthesis technology based ... https://arxiv.org/html/2104.09995v1

86. HARP: Harmonic-Aware Residual Partitioning for Neural ... https://arxiv.org/html/2607.16657v1
87. Review of end-to-end speech synthesis technology based ... https://arxiv.org/pdf/2104.09995
88. HARP: Harmonic-Aware Residual Partitioning for Neural ... https://arxiv.org/pdf/2607.16657
89. Inaudible Voice Commands - DolphinAttack https://arxiv.org/pdf/1708.09537
90. music-generation-research/README.md at master https://github.com/AI-Guru/music-generation-research/blob/master/README.md
91. AI Audio Datasets (AI-ADS) https://github.com/Yuan-ManX/ai-audio-datasets
92. sonic-pi/CHANGELOG.md at dev https://github.com/sonic-pi-net/sonic-pi/blob/dev/CHANGELOG.md
93. DRC for Movies that have quiet dialogues but very loud ... https://github.com/mpv-player/mpv/issues/8418
94. Ultimate AI Media Generation Tools Master List · GitHub https://github.com/jayeshmepani/Media-AI
95. Generating Synth Patches with AI https://gist.github.com/0xdevalias/5a06349b376d01b2a76ad27a86b08c1b
96. Plachtaa/seed-vc: zero-shot voice conversion ... https://github.com/Plachtaa/seed-vc
97. backblaze-labs/awesome-audio-generation: A curated list of AI ... https://github.com/backblaze-labs/awesome-audio-generation
98. takakhoo/Automatic-Music-Mastering-Using-Deep ... https://github.com/takakhoo/Automatic-Music-Mastering-Using-Deep-Learning
99. Truedat - Music Mood Extractor & Fingerprinter https://github.com/halrad-com/Truedat
100. A Step Towards Music Generation Foundation Model https://arxiv.org/pdf/2506.00045
101. website/content/literature/README.md at master https://github.com/sigsep/website/blob/master/content/literature/README.md
102. AnySynth: Zero-Shot Instrument Cloning via In-Context ... https://arxiv.org/html/2607.11143
103. Audio-to-Audio via Diffusion Warm Initialization https://arxiv.org/pdf/2606.18968
104. Zero-Shot Singing Voice Synthesis and Conversion with ... https://arxiv.org/pdf/2501.13870
105. A Lightweight Two-Branch Architecture for Multi-Instrument ... https://transactions.ismir.net/articles/10.5334/tismir.300

106. Assessing the Alignment of Audio Representations With ... https://ismir2025program.ismir.net/poster_283.html

107. A Review of String Instrument Synthesis Methods for Use in ... https://transactions.ismir.net/articles/10.5334/tismir.267

108. Decompile-Bench: Million-Scale Binary-Source Function ... https://proceedings.neurips.cc/paper_files/paper/2025/hash/079cf13ae174c31f148207d94d213bdc-Abstract-Datasets_and_Benchmarks_Track.html

109. MMAR: A Challenging Benchmark for Deep Reasoning in ... https://proceedings.neurips.cc/paper_files/paper/2025/file/610a7d6507d55be70c6d13d0b663227d-Paper-Datasets_and_Benchmarks_Track.pdf

110. Beethoven Symphony Excerpt Dataset (BSED): An Evaluation ... https://transactions.ismir.net/articles/10.5334/tismir.343

111. Software Tools | ISMIR https://ismir.net/resources/software-tools/

112. Open Vision Reasoner: Transferring Linguistic Cognitive ... https://neurips.cc/virtual/2025/poster/116059

113. The Catechol Benchmark: Time-series Solvent Selection ... https://proceedings.neurips.cc/paper_files/paper/2025/file/ad61e68e2c4208d5d092b3ecd6064b46-Paper-Datasets_and_Benchmarks_Track.pdf

114. V-PETL Bench: A Unified Visual Parameter-Efficient ... https://proceedings.neurips.cc/paper_files/paper/2024/hash/935de67d1a033fd517cb49d192b5c008-Abstract-Datasets_and_Benchmarks_Track.html

115. Latent Granular Resynthesis using Neural Audio Codecs https://arxiv.org/pdf/2507.19202

116. PESTO: Real-Time Pitch Estimation with Self-supervised ... https://arxiv.org/pdf/2508.01488

117. MusiQAl: A Dataset for Music Question–Answering through ... https://transactions.ismir.net/articles/10.5334/tismir.222

118. Music Arena: Live Evaluation for Text-to-Music https://proceedings.neurips.cc/paper_files/paper/2025/file/4962124728954e41e3cfe990b3a307f4-Paper-Creative_AI_Track.pdf

119. Interacting with Annotated and Synchronized Music Corpora ... https://transactions.ismir.net/articles/10.5334/tismir.212

120. ChoraleBricks: A Modular Multitrack Dataset for Wind Music ... https://transactions.ismir.net/articles/10.5334/tismir.252

121. The GigaMIDI Dataset with Features for Expressive Music ... https://transactions.ismir.net/articles/10.5334/tismir.203

122. ORIGAMISPACE: Benchmarking Multimodal LLMs in Multi- ... https://proceedings.neurips.cc/paper_files/paper/2025/file/d2a7441f07da2c504b280c6bb583709b-Paper-Conference.pdf
123. Interacting with Annotated and Synchronized ... https://transactions.ismir.net/articles/212/files/68720942a2a8e.pdf
124. Comprehensive Evaluation of Safety and Adversarial ... https://papers.neurips.cc/paper_files/paper/2025/file/1ca47465a87b8e125d9076b2e6ac6c96-Paper-Datasets_and_Benchmarks_Track.pdf
125. NeurIPS 2025 Schedule https://neurips.cc/virtual/2025/calendar
126. arXiv_daily/daily_papers/20260106_Tue/audio.md at main https://github.com/advanced-cs/arXiv_daily/blob/main/daily_papers/20260106_Tue/audio.md
127. OpenReviewExplorer/OpenReview scraping.ipynb at master https://github.com/Chillee/OpenReviewExplorer/blob/master/OpenReview%20scraping.ipynb
128. A Token-based Neural Synthesizer for Instrument Cloning ... https://ieeexplore.ieee.org/document/10888403/
129. Jin Woo Lee (@szin94) / Posts / X https://x.com/szin94